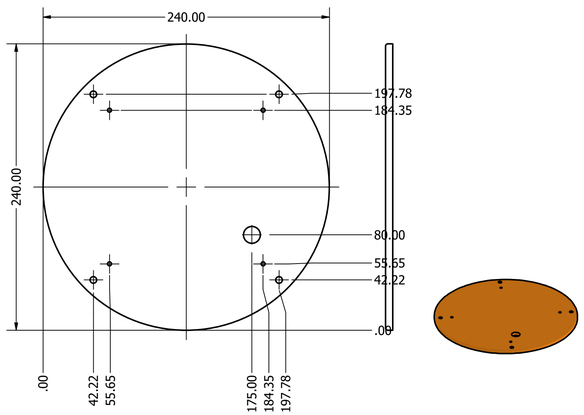
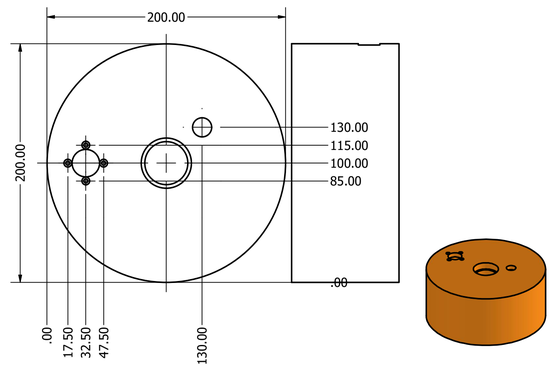
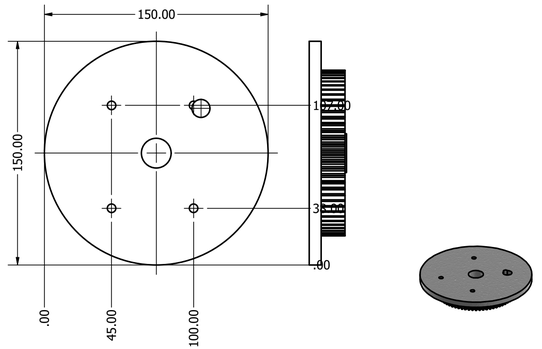
RA-101 BASEPLATE
PETG, QTY 1, 341 g (INVENTOR)
SCALE 1:2.5



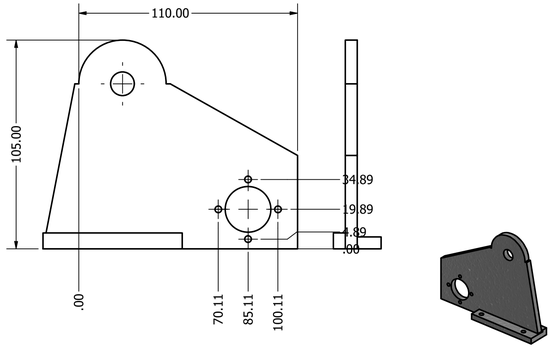
RA-102 BASEHOUSING
PETG, QTY 1, 611 g (INVENTOR)
SCALE 1:2.5



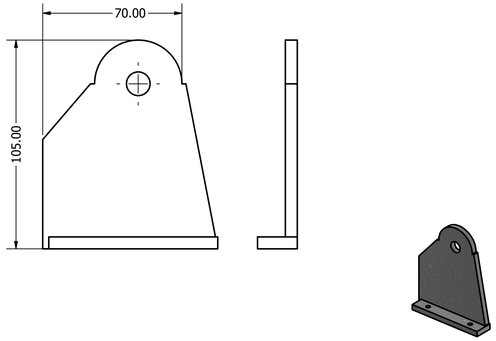
RA-103 TURNTABLE
PA6-CF, QTY 1, 238 g (INVENTOR)
SCALE 1:2



RA-104 SHOULDERCHEEKR
PA6-CF, QTY 1, 68 g (INVENTOR)
SCALE 1:1.5



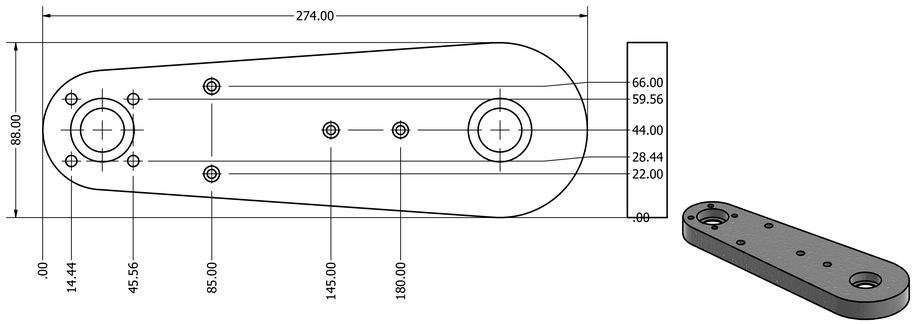
RA-105 SHOULDERCHEEKL
PA6-CF, QTY 1, 53 g (INVENTOR)
SCALE 1:1.5



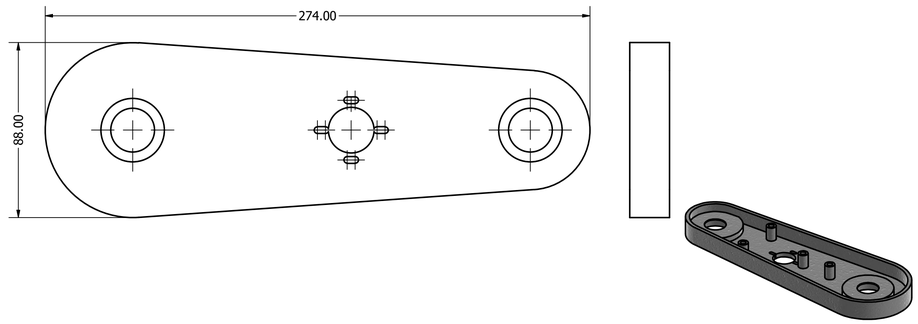
NOTES:
1. MATERIAL: PA6-CF (BAMBU LAB) UNLESS NOTED; PETG WHERE NOTED. DRY PA6-CF 80 C / 8 H.
2. PRINT: 0.4 MM NOZZLE, 0.2 MM LAYERS, 5 WALLS, 40% GYROID, 6 TOP/BOTTOM. ANNEAL PA6-CF 80 C / 6 H.
3. BEARING SEATS PRINTED -0.10 MM, PRESS FIT AFTER TEST PRINT. M3/M4 BRASS HEAT-SET INSERTS WHERE 4.0 / 5.6 HOLES.
4. GEARS: 20 DEG PRESSURE ANGLE, PRINT FLAT (TEETH IN XY). UNTOLERANCED DIMS +/-0.2 MM.
5. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts.py (PARAMETRIC SCRIPT).

DRAWN Jay Patel	10/28/2026	
CHECKED		TITLE: RoboticArm6DOF - printed part manufacturing drawings
QA		
MFG		
APPROVED		
		SIZE D | DWG NO RA-DWG-100 | REV
		SCALE 0.40 : 1 | SHEET 1 OF 4

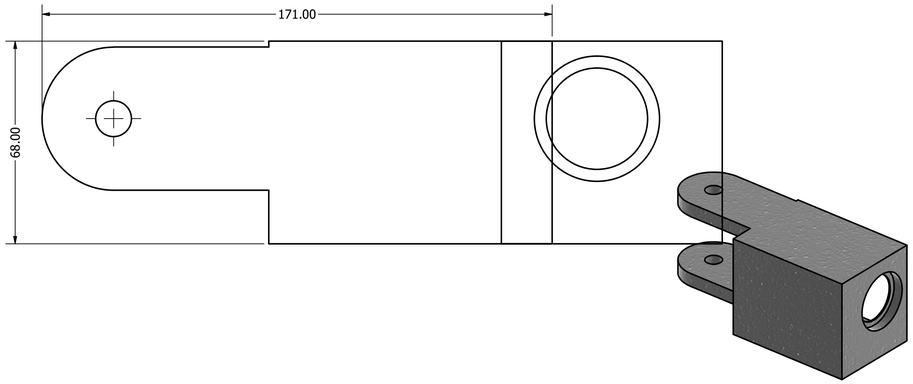
RA-106 UPPERARMA
PA6-CF, QTY 1, 158 g (INVENTOR)
SCALE 1:1.5



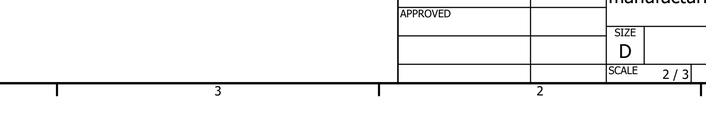
RA-107 UPPERARMB
PA6-CF, QTY 1, 157 g (INVENTOR)
SCALE 1:1.5



RA-108 FOREARM
PA6-CF, QTY 1, 202 g (INVENTOR)
SCALE 1:1



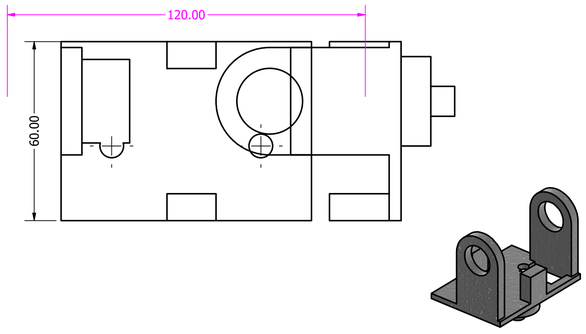
RA-109 FOREARMCOVER
PETG, QTY 1, 25 g (INVENTOR)
SCALE 1:1



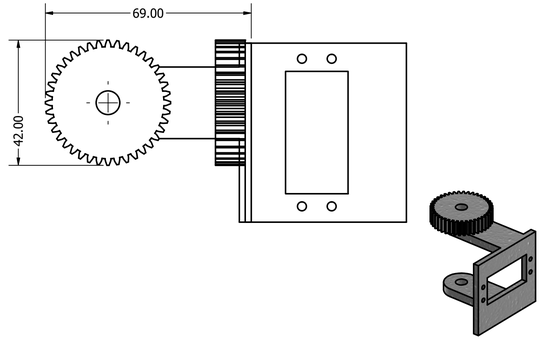
NOTES:
1. MATERIAL: PA6-CF (BAMBU LAB) UNLESS NOTED; PETG WHERE NOTED. DRY PA6-CF 80 C / 8 H.
2. PRINT: 0.4 MM NOZZLE, 0.2 MM LAYERS, 5 WALLS, 40% GYROID, 6 TOP/BOTTOM. ANNEAL PA6-CF 80 C / 6 H.
3. BEARING SEATS PRINTED -0.10 MM, PRESS FIT AFTER TEST PRINT. M3/M4 BRASS HEAT-SET INSERTS WHERE 4.0 / 5.6 HOLES.
4. GEARS: 20 DEG PRESSURE ANGLE, PRINT FLAT (TEETH IN XY). UNTOLERANCED DIMS +/-0.2 MM.
5. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts.py (PARAMETRIC SCRIPT).

DRAWN Jay Patel	10/28/2026	
CHECKED		
QA		TITLE: RoboticArm6DOF - printed part manufacturing drawings
MFG		
APPROVED		
		SIZE D · DWG NO RA-DWG-100 · REV
		SCALE 2 / 3 · SHEET 2 OF 4

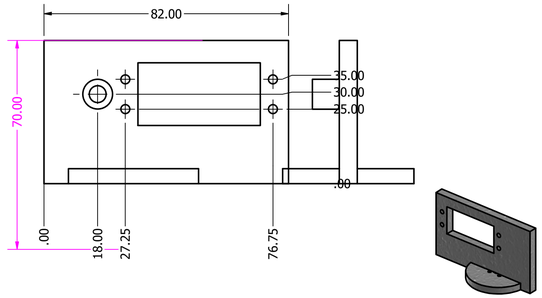
RA-110 WRISTYOKE
PA6-CF, QTY 1, 58 g (INVENTOR)
SCALE 1:1



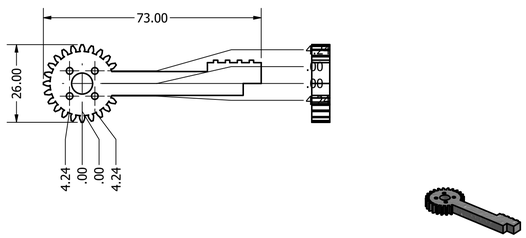
RA-111 WRISTBRACKET
PA6-CF, QTY 1, 39 g (INVENTOR)
SCALE 1:1



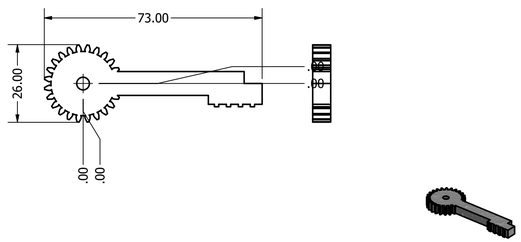
RA-112 GRIPPERBASE
PA6-CF, QTY 1, 27 g (INVENTOR)
SCALE 1:1



RA-113 FINGERL
PA6-CF, QTY 1, 5 g (INVENTOR)
SCALE 1:1



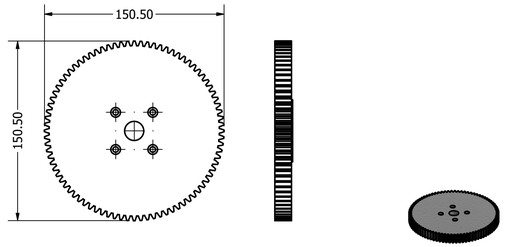
RA-114 FINGERR
PA6-CF, QTY 1, 6 g (INVENTOR)
SCALE 1:1



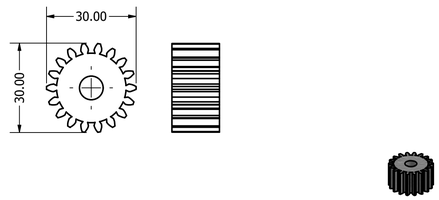
NOTES:
1. MATERIAL: PA6-CF (BAMBU LAB) UNLESS NOTED; PETG WHERE NOTED. DRY PA6-CF 80 C / 8 H.
2. PRINT: 0.4 MM NOZZLE, 0.2 MM LAYERS, 5 WALLS, 40% GYROID, 6 TOP/BOTTOM. ANNEAL PA6-CF 80 C / 6 H.
3. BEARING SEATS PRINTED -0.10 MM, PRESS FIT AFTER TEST PRINT. M3/M4 BRASS HEAT-SET INSERTS WHERE 4.0 / 5.6 HOLES.
4. GEARS: 20 DEG PRESSURE ANGLE, PRINT FLAT (TEETH IN XY). UNTOLERANCED DIMS +/-0.2 MM.
5. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts.py (PARAMETRIC SCRIPT).

DRAWN Jay Patel	10/28/2026	
CHECKED		TITLE: RoboticArm6DOF - printed part manufacturing drawings
QA		
MFG		
APPROVED		
		SIZE D · DWG NO RA-DWG-100 · REV
		SCALE 1 : 1 · SHEET 3 OF 4

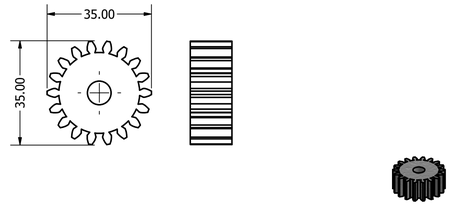
RA-203 J2GEAR84T
PA6-CF, QTY 1, 171 g (INVENTOR)
SCALE 1:2.5



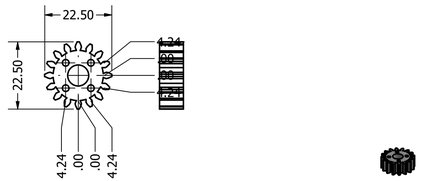
RA-204 J1PINION18T
PA6-CF, QTY 1, 9 g (INVENTOR)
SCALE 1:1



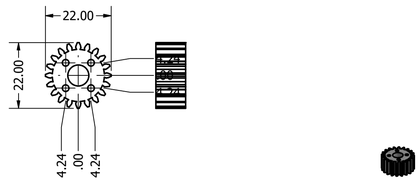
RA-205 J2PINION18T
PA6-CF, QTY 1, 11 g (INVENTOR)
SCALE 1:1



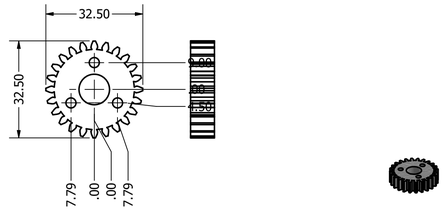
RA-206 J4PINION
PA6-CF, QTY 1, 2 g (INVENTOR)
SCALE 1:1



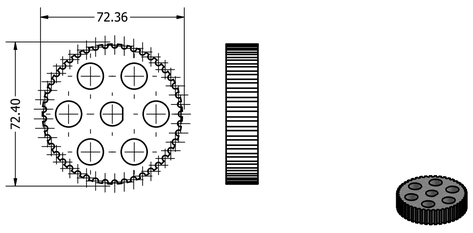
RA-207 J5PINION
PA6-CF, QTY 1, 3 g (INVENTOR)
SCALE 1:1



RA-208 J4GEAR
PA6-CF, QTY 1, 5 g (INVENTOR)
SCALE 1:1



RA-209 PULLEYHTD5M_45T
PA6-CF, QTY 1, 48 g (INVENTOR)
SCALE 1:1.5



NOTES:
1. MATERIAL: PA6-CF (BAMBU LAB) UNLESS NOTED; PETG WHERE NOTED. DRY PA6-CF 80 C / 8 H.
2. PRINT: 0.4 MM NOZZLE, 0.2 MM LAYERS, 5 WALLS, 40% GYROID, 6 TOP/BOTTOM. ANNEAL PA6-CF 80 C / 6 H.
3. BEARING SEATS PRINTED -0.10 MM, PRESS FIT AFTER TEST PRINT. M3/M4 BRASS HEAT-SET INSERTS WHERE 4.0 / 5.6 HOLES.
4. GEARS: 20 DEG PRESSURE ANGLE, PRINT FLAT (TEETH IN XY). UNTOLERANCED DIMS +/-0.2 MM.
5. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts.py (PARAMETRIC SCRIPT).

GEAR DATA (20 DEG PRESSURE ANGLE, STANDARD ADDENDUM, PRINTED PA6-CF)
J1: m1.5 18T:72T RATIO 4.000 CENTRE 67.50 FACE 16 PCD 27.00 / 108.00
J2: m1.75 18T:84T RATIO 4.667 CENTRE 89.25 FACE 14 PCD 31.50 / 147.00
J4: m1.25 16T:24T RATIO 1.500 CENTRE 25.00 FACE 8 PCD 20.00 / 30.00
J5: m1 20T:40T RATIO 2.000 CENTRE 30.00 FACE 10 PCD 20.00 / 40.00
GRIP: m1 24T:24T RATIO 1.000 CENTRE 24.00 FACE 6 PCD 24.00 / 24.00
J3: HTD 5M 15T:45T BELT 375-5M-15, CENTRE 109.91 (4 MM TENSION SLOTS)
J1 LEWIS: PINION 17.8 MPA, GEAR 12.7 MPA (ALLOW 30)
J2 LEWIS: PINION 11.8 MPA, GEAR 8.3 MPA (ALLOW 30)
J4 LEWIS: PINION 18.9 MPA, GEAR 16.6 MPA (ALLOW 30)
J5 LEWIS: PINION 13.2 MPA, GEAR 10.9 MPA (ALLOW 30)

DRAWN Jay Patel	10/28/2026	TITLE: RoboticArm6DOF - printed part manufacturing drawings
CHECKED		
QA		
MFG		
APPROVED		
		SIZE D · DWG NO RA-DWG-100 · REV
		SCALE 0.40 : 1 · SHEET 4 OF 4